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GeoRoute: Geometry-Aware Hybrid Inference for Traffic Future-Frame Prediction

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Abstract. Long-horizon future-frame prediction is important for autonomous driving, traffic surveillance, and intelligent transportation systems, yet remains challenging due to temporal ghosting, geometry drift, and inconsistent object motion. Recent latent video diffusion models have achieved impressive visual quality, but directly applying them to structured traffic scenes often leads to unstable geometry and degraded temporal coherence over extended horizons. We present a training-free inference framework that stabilizes reliable static structure in pretrained video predictions through multi-frame temporal context and view conditioned routing. For front-camera videos, our method refines generated futures with a multi-frame depth-based point-splatting renderer that projects static content from observed history frames while preserving dynamic regions from the generative base model. For heterogeneous traffic views, a frozen vision-language model infers a coarse camera group from the observed clip and selects a specialized motion-based predictor. The framework requires neither retraining nor fine-tuning of the underlying video model and can be applied directly to pretrained generators. We validate the proposed framework on the AI City Challenge Track 5 benchmark [36], where our final system ranks fifth on the official full-test leaderboard. The leaderboard results show competitive overall low-level fidelity, while qualitative comparisons support the intended static-structure effect. The framework does not change the original generator architecture. Our code will be available on Github.

Keywords: Future-Frame Prediction, Traffic Video Forecasting, Video Diffusion Models, Training-Free Inference, Geometry-Aware Refinement, Temporal Consistency

1 Introduction

Future-frame prediction is important for intelligent transportation systems, autonomous driving, and traffic surveillance. Given a short observed video clip, the goal is to predict future frames that preserve scene geometry, object motion, and temporal coherence. This problem has been studied through recurrent, predictive-coding, motion-aware, and flow-based video prediction methods [35,26,41,39,25,24,8,11]. In traffic scenarios, the task is especially challenging due to ego-motion, independently moving vehicles and pedestrians, occlusions, and diverse camera viewpoints [45,19].

Recent latent video diffusion models have greatly improved visual generation quality by extending latent diffusion with temporal layers, spatiotemporal attention, or transformer backbones [32,17,40,3,2,6,13,14,20]. However, directly applying pretrained video generators to long-horizon traffic forecasting often produces temporal ghosting, afterimages, geometry drift, and inconsistent object motion. Training-free video generation and editing methods also face similar frame-to-frame consistency challenges [18,44,12,28].

Fine-tuning a large video generator on the target benchmark is expensive, data-sensitive, and difficult under challenge constraints. Training-free diffusion control methods adapt pretrained models by manipulating attention, propagating features, or enforcing cross-frame consistency [15,4,5,18,12], but they do not directly address the static-geometry artifacts that appear in traffic future-frame prediction. Geometry and motion cues offer complementary signals: optical flow can propagate visual content [25,37], while monocular depth and semantic segmentation help separate static scene structure from dynamic actors [31,30,7].

We propose a training-free inference framework for long-horizon traffic future-frame prediction. For front-camera driving videos, we use a pretrained video generator as an appearance and dynamic-content prior, then refine static regions with a multi-frame depth-based point-splatting renderer. Depth and actor masks are estimated from observed history frames, while history-to-generated-view alignment is obtained by matching static features between an observation and each generated base frame. Reliable static pixels are then projected using z-buffer splatting. Dynamic or low-confidence regions are preserved from the generative base prediction. Thus, the geometry branch stabilizes static structure rather than re-estimating future actor trajectories. For other visual regimes, we use specialized motion-based predictors, since elevated or nearly fixed views often favor deterministic propagation over generic front-camera generation.

Our framework operates entirely at inference time and requires neither retraining nor fine-tuning of the underlying video generator. We evaluate on AI City Challenge Track 5, using BDD front-camera videos [45] and WTS traffic views [19]. The benchmark reports PSNR and SSIM [42] for reconstruction fidelity, LPIPS [46] for perceptual similarity, CLIP similarity [29] for semantic alignment, and FID/FVD [16,38] for image and video distribution quality. Our final system achieves a challenge score of **73.28**.

Our contributions are summarized as follows:

- We propose a training-free inference framework for long-horizon traffic future-frame prediction that improves static-geometry stability and low-level structural fidelity without modifying model parameters or sampling schedules.

- We introduce a confidence-aware geometry refinement module with multi-frame history support. The module projects reliable static pixels from observed frames into future views, while preserving dynamic and low-confidence regions from the generative base prediction.
- We design a Qwen-assisted view-conditioned routing strategy that combines front-camera generative refinement with view-specific motion-based predictors.
- We validate the framework on AI City Challenge Track 5 [36], achieving a final challenge score of **73.28**.

2 Related Work

2.1 Latent Video Diffusion Models

Diffusion models have become a dominant paradigm for visual generation, while latent diffusion reduces the cost of pixel-space denoising by operating in a compressed representation [32]. Video Diffusion Models and MCVD extended diffusion to temporal generation and conditional future-frame prediction [17,40]. Subsequent latent video models introduced temporal U-Net modules, latent-space processing, and image–video joint training to improve resolution and efficiency [3,6,2,13]. More recent systems such as HunyuanVideo and LTX-Video employ transformer-based spatiotemporal architectures for scalable video synthesis and stronger conditioning alignment [20,14]. Despite these advances, inference-only application to long-horizon traffic prediction can still produce temporal ghosting, geometry drift, and inconsistent object motion.

2.2 Training-Free Diffusion Control and Video Editing

Training-free diffusion control reuses representations within pretrained models without updating their parameters. Prompt-to-Prompt manipulates cross-attention maps, while MasaCtrl shares mutual self-attention features to preserve spatial layout and appearance [15,4]. For videos, Text2Video-Zero introduces cross-frame attention and latent motion dynamics without video-model training [18]. Pix2Video, Rerender-A-Video, FateZero, TokenFlow, and VidToMe propagate attention, diffusion features, correspondences, or tokens across frames to improve editing consistency [5,44,28,12,22]. These approaches primarily target text-guided generation or editing. GeoRoute instead refines future predictions with external geometry recovered from the observed history and leaves the pretrained sampling process unchanged.

2.3 Temporal Consistency and Geometry-Guided Refinement

Optical flow provides dense temporal correspondences and is widely used for warping or propagating visual content. RAFT estimates flow through recurrent all-pairs matching, while FlowVid combines flow guidance with spatial conditions to reduce errors caused by imperfect correspondence [37,23]. Nevertheless, flow becomes unreliable under occlusion, disocclusion, rapid camera motion, and independently moving actors.

Geometry-based view synthesis provides a complementary way to preserve scene structure. SynSin projects predicted 3D features into novel views, layered depth representations explicitly model occluded scene content, and splatting operators resolve multiple source samples that map to the same target location [43,34,27]. Inspired by these principles, GeoRoute projects only masked static pixels from multiple observed frames, resolves visibility with a z-buffer, and uses confidence-aware blending to preserve generated content where geometry is unreliable.

2.4 Hybrid Prediction for Heterogeneous Traffic Views

Traffic datasets contain substantially different camera configurations. BDD100K primarily provides front-facing driving videos with ego-motion, whereas WTS includes vehicle, fixed, and overhead viewpoints [45,19]. Classical optical flow and adaptive background models remain useful when camera geometry is stable [9,47]. A single predictor may therefore be suboptimal across all views. GeoRoute uses Qwen-assisted view-conditioned routing: front-camera videos receive generative prediction followed by geometric refinement, while traffic-camera views use conservative motion propagation.

3 Method

3.1 Problem Formulation

Let x_t denote the RGB frame observed at time t. Given a history of T frames, $\{x_1, \ldots, x_T\}$, our goal is to predict the next K faicity2026track5 $\{\hat{y}_1, \ldots, \hat{y}_K\}$. Thus, x_T is the last observation and $\hat{y}_j$ is the prediction at future step j.

Traffic prediction requires two different capabilities. Static structures such as roads, lane markings, curbs, and buildings should remain geometrically stable, whereas vehicles, pedestrians, and newly visible regions require plausible synthesis. These requirements become harder under ego-motion, occlusion, and heterogeneous camera viewpoints. GeoRoute therefore separates the problem into two responsibilities: a pretrained generator proposes future appearance and dynamic content, while an explicit geometry module corrects only static regions for which observed evidence is reliable. Views with nearly stable camera geometry use a simpler motion-propagation branch instead.

3.2 Framework Overview

Figure 1 shows the complete data flow. GeoRoute receives an observed clip and its available textual description. A frozen Qwen2.5-VL [1] model first classifies the observed clip into a coarse view group and selects the corresponding prediction branch. Front-camera driving videos pass through prompt construction, LTX-Video generation [14], and geometry-aware refinement. Traffic-camera videos pass through one of two motion-based predictors according to the predicted view group.

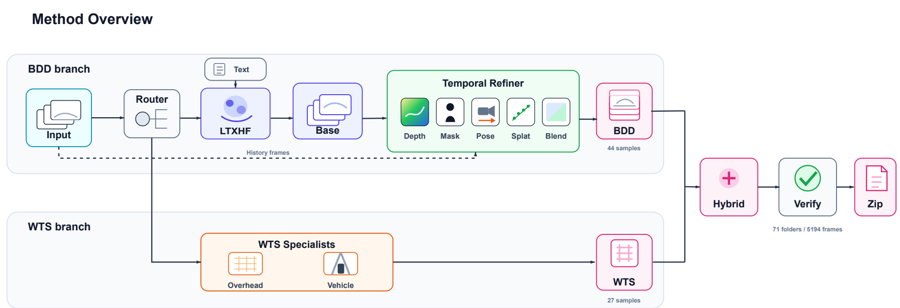

Fig. 1: **GeoRoute overview.** Qwen infers a coarse view group from the observed clip and routes it to either the text-conditioned LTX-Video [14] and geometry-refinement branch or a stable-view motion predictor. Each branch returns the same sequence of future frames.

This design follows the physical behavior of the input rather than assuming that one predictor is optimal for every camera. Front-camera videos contain strong perspective change and disocclusion, so a generator is needed for content that cannot be copied from the past. However, its output may contain ghosted lane boundaries or drifting buildings. GeoRoute anchors those static regions to real observations. For overhead or slowly moving traffic cameras, the observed background is already a strong future reference; deterministic propagation avoids unnecessary generative drift. All routing rules are fixed before evaluation, and no future ground-truth frame is used.

3.3 View-Conditioned Routing

The frozen Qwen2.5-VL-7B-Instruct [1] router independently matches the first, middle, and last observed frames of each video to one of three operator-oriented visual-regime prototypes: (i) forward-driving views, (ii) stable elevated or fixed views, and (iii) other oblique roadside or mobile views. They select, respectively, generative refinement, median-background actor propagation, or conservative region-normalized flow propagation. Routing therefore depends on visual behavior rather than dataset identity or physical camera protocol.

Following the camera patterns represented by BDD100K and WTS [45,19], the instruction uses observable cues: viewpoint and road geometry, cross-frame background displacement, and whether motion is global or actor-local. A low forward-facing viewpoint with a road vanishing point and scene-wide background displacement indicates front-camera driving. An elevated viewpoint with a nearly stationary background and actor-local motion indicates an overhead/fixed camera; remaining oblique roadside or mobile views are assigned to the third regime.

Qwen [1] receives no dataset identity, metadata, file name, target frame, or manual label. The prototypes, prompt, parser, and branch mapping are fixed for all samples, so clips with similar viewpoint and motion structure receive the same route. This is label-free zero-shot routing; no clustering model is fitted on the test set. An unparsable response falls back to the conservative third branch. Figure 2 shows representative unlabeled inputs.

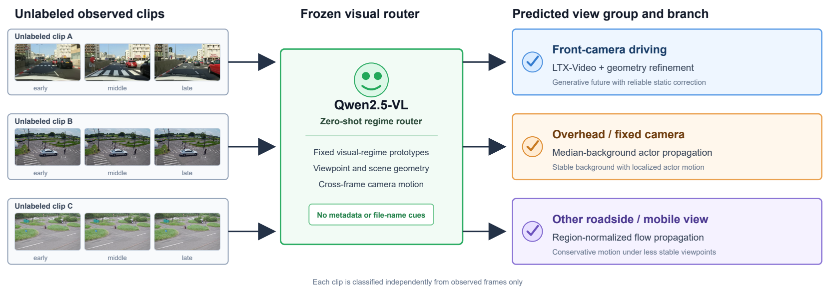

Fig. 2: **Qwen-assisted view routing.** Three observed frames are matched to fixed visual-regime prototypes; the selected regime determines the corresponding inference branch.

3.4 Qwen-Assisted Prompt and Base Prediction

For a front-camera sample, the base generator is conditioned on both the observed clip and a compact prompt. Frozen Qwen2.5-VL-7B-Instruct [1] summarizes sparsely sampled history frames together with the supplied text and predicted view group [1]. Its structured output is restricted to visible scene layout, actor categories, and coarse motion context; it does not infer exact future trajectories. Sampling across the history reduces the influence of transient occlusion.

Figure 3 illustrates this decomposition. The view, scene, actor, and motion fields are inserted into a deterministic template. A short negative field discourages common generation artifacts such as temporal ghosting, duplicated actors, warped road markings, flicker, and abrupt viewpoint changes. Empty fields in the fixed schema fall back to generic traffic-scene phrases rather than invented details.

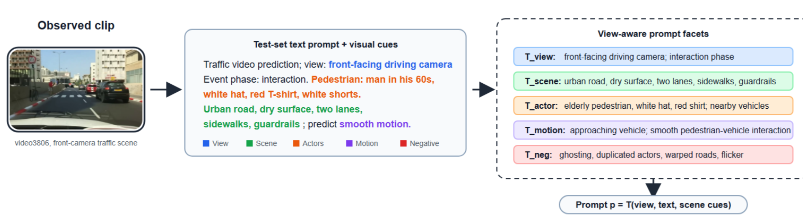

Fig. 3: **Qwen-assisted prompt construction.** Observed frames and the available sample text are summarized into grounded view, scene, actor, and motion cues, then assembled by a fixed template.

Qwen remains frozen and is used only at inference time. Its view output selects the branch, while its scene facets condition pretrained LTX-Video [14] within the front-camera branch; geometry refinement does not use language predictions. Let b_j be the generated frame at future step j. It provides the default full-frame prediction, including dynamic and disoccluded content that cannot be projected from history. Later modules replace only confident static regions and otherwise retain b_j. GeoRoute therefore claims improved static-geometry stability and low-level structural fidelity, not explicit recovery of actor trajectories.

3.5 Geometry-Aware Static Refinement

Figure 4 expands the refinement stage. Its purpose is simple: recover reliable background pixels from real history frames, move them to the predicted camera view, and leave all other pixels to the generator.

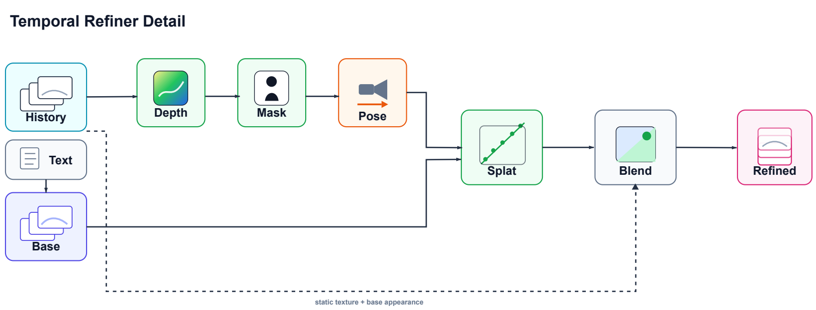

Fig. 4: **Geometry-aware static refinement.** Depth, actor masks, and camera motion turn observed history frames into a rendered static layer. Confidence-aware blending applies this layer only where the projection is reliable.

Selecting static observations. We select up to eight history frames, ordered from old to recent. The most recent frame is usually the closest visual reference, while older frames may contain background regions that later become disoccluded. For each selected frame, DPT Hybrid-MiDaS estimates relative depth [31,30], and DeepLabV3-ResNet50 estimates an actor mask [7]. Person, bicycle, motorcycle, car, bus, and related vehicle classes are excluded from static projection. We also mask actors in each base frame so that a projected road or building cannot overwrite a generated vehicle.

The two masks have different roles: the history mask prevents past actors from being copied to outdated positions, while the base-frame mask prevents projected background from overwriting actors generated for the future.

Depth is used only as relative geometry. The predicted inverse depth is normalized with robust image percentiles and converted to a bounded positive range. Because calibrated intrinsics are unavailable, we use an approximate pinhole camera whose principal point is the image center. These approximations are sufficient for conservative local projection, but they are not interpreted as metric 3D reconstruction.

Aligning a history frame to a future view. For every future base frame b_j, ORB features are matched between static regions [33]. Matches on actors are discarded. The remaining points, together with history depth, are used by EPnP-RANSAC [21,10] to estimate the camera transform. To make the geometric operation explicit, a static pixel u with depth $D(u)$ is mapped to the future view as

$$\tilde{u}' \sim K\big(R\big(D(u)K^{-1}\tilde{u}\big) + t\big)\,.$$

Here, $\tilde{u}$ and $\tilde{u}'$ are the pixel coordinates before and after projection, K is the approximate camera matrix, and (R, t) is the estimated relative pose. This equation is used only for static pixels. A pose is rejected when there are fewer than 20 ORB matches or 12 RANSAC inliers. These rejected raw estimates may be filled from the pose sequence before the method falls back to b_j.

Specifically, interpolation concerns the transform from the most recent history frame to each future base frame. When a raw pose is missing and at least two future steps have valid poses, rotation is represented by its three-component Rodrigues vector and concatenated with the three translation components. Each of the six components is linearly interpolated over the future-step indices, with the nearest valid pose extended at sequence boundaries, and the resulting sequence is smoothed by a triangular window of length seven. The implementation does not impose an additional maximum-gap threshold; instead, every filled pose is down-weighted to $Q_j = 0.35$. If fewer than two valid poses are available, no interpolation is performed and a missing step retains the base frame b_j.

Older history frames are first aligned to the most recent history frame, then their transforms are composed with the most-recent-to-future transform. This puts all selected observations in one future coordinate system without fitting an unrelated pose for every history–future pair.

Rendering and multi-frame fusion. Each aligned history frame produces a candidate background layer. When several source pixels land at the same target location, z-buffer splatting keeps the closest point [43,27]. Candidate layers are then combined from old to recent. A recent observation replaces an older one where both cover the same pixel; an older observation is used only to fill a location that newer frames do not cover. The resulting rendered background for future step j is denoted by r_j. Pixels with no projected observation are copied from b_j, so the renderer never creates an empty hole.

This recent-first rule is deliberately conservative. It avoids averaging slightly misaligned projections and gives older frames one clear role: expanding coverage in static regions that are absent from the latest observation.

3.6 Confidence and Final Blending

Projection alone is not reliable enough to replace the generated image. Depth may be noisy, pose may be weak, and projected colors may disagree with the future appearance. We therefore compute a confidence value $w_j(u)$ for every pixel u. Its four factors are shown directly in the equation:

$$w_j(u) = \underbrace{C_j(u)}_{\text{covered}}\underbrace{\big(1 - A_j(u)\big)}_{\text{static}}\underbrace{\exp(-\Delta_j(u)/32)}_{\text{color agreement}}\underbrace{Q_j}_{\text{pose quality}}.$$

$C_j(u)$ equals one when at least one history frame covers the pixel and zero otherwise. $A_j(u)$ is the actor mask of the base frame. $\Delta_j(u)$ is the mean absolute RGB difference between $r_j(u)$ and $b_j(u)$ on the 8-bit color scale. Finally, Q_j is 1 for an accepted pose, 0.35 for a pose obtained by temporal interpolation, and 0 when no pose is available. Thus, a pixel receives high confidence only when it is covered, static, photometrically compatible, and supported by a usable camera transform.

Coverage rejects rendering holes, the actor term protects dynamic content, color agreement suppresses misaligned projections, and pose quality rejects unsupported camera transforms. Their product is conservative because one weak factor is sufficient to reduce the refinement weight.

We erode and blur this map to remove isolated projection boundaries, then apply an exponential moving average across future steps to reduce flicker. The final actor mask is applied once more after smoothing to prevent confidence from leaking onto moving objects.

The refined prediction is a convex blend of the generated base frame and the rendered static layer:

$$\hat{y}_j(u) = \big(1 - \alpha w_j(u)\big)b_j(u) + \alpha w_j(u) r_j(u), \qquad \alpha = 0.75.$$

This expression has a direct interpretation. If $w_j(u) = 0$, the output is the LTX-Video prediction [14]. If confidence increases, more observed static content is used, up to the maximum strength α. The method therefore stabilizes roads, curbs, lane markings, and buildings without claiming to correct an incorrect actor trajectory already generated in b_j.

The blend remains soft even at high confidence. Retaining part of b_j reduces small exposure and color differences between observed and generated content, while gradual confidence boundaries avoid visible seams around lane markings, curbs, and actor masks. This is more stable than hard replacement, which can turn a small projection error into a sharp spatial artifact.

3.7 Motion Prediction for Stable Views

Traffic-camera views use observed image motion instead of LTX-Video generation [14]. Both predictors estimate Farneback optical flow [9] between the last two observed frames and extrapolate it with a decaying displacement. The two variants differ in what they preserve.

For overhead and fixed cameras, a per-pixel temporal median provides a stable background. Flow magnitude and deviation from this median identify compact moving regions. These actor regions are propagated forward and composited over the unchanged median background. Large connected regions are rejected because they are more likely to represent illumination or global camera change than a single actor.

For vehicle-mounted and IP-camera views, global motion is more common, so a hard actor/background decomposition is less reliable. We instead convert flow magnitude into a soft regional mask. The last observed frame is warped by an accumulated, clipped, and decayed flow, then blended with the unwarped frame through this mask. This preserves local motion while limiting long-horizon distortion. Exact thresholds, decay factors, and displacement limits are given in Sec. 4.1.

Neither stable-view predictor learns object trajectories. They extrapolate observed image-space motion under a conservative background prior and are used only for camera patterns where that assumption is appropriate.

3.8 Inference Procedure

Algorithm 1 summarizes the method without introducing additional notation.

Algorithm 1 Training-free GeoRoute inference

Require: Observed clip and available text
Ensure: Predicted future frames
Classify the observed clip with frozen Qwen2.5-VL-7B-Instruct
if the sample is a front-camera driving view **then**
  Build a grounded prompt with frozen Qwen2.5-VL-7B-Instruct
  Generate base future frames with LTX-Video
  Estimate history depth and actor masks
  Estimate the recent-history-to-future pose sequence
  Interpolate and smooth missing poses when at least two are valid
  for each future step $j = 1, \ldots, K$ **do**
    if a direct or filled pose is available **then**
      Render and fuse the multi-frame static layer r_j
      Compute the projection confidence w_j
      Blend r_j with the base frame b_j to obtain $\hat{y}_j$
    else
      Keep the base prediction $\hat{y}_j = b_j$
    end if
  end for
else if the sample is an overhead or fixed traffic-camera view **then**
  Apply median-background actor-layer propagation
else
  Apply conservative region-normalized flow propagation
end if
return The predicted future sequence

The complete pipeline is training-free after the pretrained components are loaded. It uses only the observed clip and available text; no target future frame is used during routing, prompting, rendering, or blending.

4 Experiments

4.1 Experimental Setup

We evaluate GeoRoute on the AI City Challenge Track 5 benchmark [36] long-horizon traffic future-frame prediction. According to the benchmark annotations, the test set contains 71 videos and 5194 target frames: 44 front-camera videos, 7 overhead/fixed-camera videos, and 20 vehicle/IP-camera videos. These labels and counts are reported only to describe the evaluation set; they are not supplied to the router. At inference, Qwen [1] classifies every observed clip by matching it to one of three fixed visual-regime prototypes; the selected regime determines the corresponding prediction branch. A post-hoc audit against the benchmark-provided reference groups found agreement for all 71 videos.

Ground-truth future frames for the official test set are not publicly available, so test metrics are obtained from the challenge evaluation server. All routing rules and hyperparameters are fixed across the test set; no target frames or per-video parameter tuning are used during inference.

All experiments are run on one NVIDIA RTX A6000 GPU with 48GB memory, and full-test inference finishes within one day. The framework is training-free: no generator weights, adapters, or diffusion sampling schedules are updated. Table 4.1 summarizes the main implementation settings.

Component	Setting
BDD generator	Distilled LTX-Video 2B; 1216 × 704 resized to 1280 × 720; 30 FPS; seed 17; 8 steps; guidance 1.0; STG 0; BF16
Video conditioning	History capped at 41 frames; sequence lengths adjusted to $8n+1$
Text conditioning	Frozen Qwen2.5-VL-7B-Instruct; structured scene summary from observed frames; deterministic prompt template
View routing	Frozen Qwen2.5-VL-7B-Instruct; first/middle/last observed frames; fixed three-way output schema; conservative vehicle/IP fallback
Geometry refinement	Up to 8 history frames, stride 2; point stride 2; splat radius 1; $\alpha = 0.75$; focal scale 0.9
Pose and confidence	Min. 20 ORB matches and 12 PnP-RANSAC inliers; smoothing window 7; photometric scale 32; confidence EMA 0.65
WTS optical flow	Farneback at maximum width 426 with OpenCV parameters $(0.5, 3, 15, 3, 5, 1.2, 0)$
WTS overhead/fixed	Median-background actor propagation; flow threshold 0.35; RGB thresholds 7/22; component area 20 pixels–2.5%; blur $\sigma = 1.0$; decay 0.985; maximum scale 12.0
WTS vehicle/IP	Region-normalized propagation; threshold 1.5; temperature 0.75; blur $\sigma = 3.0$; decay 0.96; maximum scale 6.0

4.2 Main Results

Table 1: Official AI City Challenge Track 5 full-test results.

Rank	Team	Final	PSNR	SSIM	LPIPS	CLIP	FID	FVD
1	Qyn	76.4866	20.1202	0.6503	0.2463	0.9498	22.4089	21.7907
2	SSUPER	76.0385	19.7271	0.6300	0.2487	0.9384	21.1641	19.4627
3	Latent Painter	75.1297	19.7213	0.6504	0.2661	0.9452	26.5176	24.7875
4	CHTTL_A30	74.0544	18.8573	0.5970	0.2814	0.9423	23.7849	24.1105
5	VGU_ai_lab	**73.2843**	**19.7360**	**0.6472**	**0.2942**	**0.9454**	**33.6081**	**29.3483**

Table 1 shows that GeoRoute ranks 5th on the official full-test leaderboard with a final score of 73.2843. Among the top five teams, our method achieves the second-highest PSNR and an SSIM close to the best reported value. These results indicate strong preservation of scene layout, road structure, and low-level appearance, consistent with the goal of anchoring reliable static regions to observed history frames.

The remaining gap mainly appears in LPIPS, FID, and FVD, which are more sensitive to perceptual realism and video-level distribution matching. Dynamic actors, disoccluded regions, and newly synthesized content are deliberately preserved from the frozen base generator and remain difficult to correct through static geometry refinement alone. Further improvements therefore require stronger dynamic-object priors and task-specific video realism.

4.3 Ablation Study

We incrementally add prompt conditioning, geometry refinement, multi-frame history, confidence-aware blending, and Qwen-assisted [1] routing to the base predictor. The table summarizes challenge-server development trends rather than an independent validation-set ablation.

Table 2: Cumulative ablation of the proposed components.

Configuration	Base	Prompt	Geo.	Hist.	Conf.	Route	Final	PSNR	SSIM	LPIPS↓	CLIP↑	FID↓	FVD↓
Base prediction	✓						66.8	18.21	0.603	0.356	0.930	55.7	37.9
Prompt conditioning	✓	✓					67.2	18.29	0.608	0.350	0.934	53.6	36.8
Static geometry	✓	✓	✓				69.2	18.59	0.622	0.331	0.940	47.8	31.3
Multi-frame history	✓	✓	✓	✓			69.4	18.61	0.620	0.331	0.940	47.2	31.2
Confidence blending	✓	✓	✓	✓	✓		71.6	19.10	0.636	0.310	0.943	39.8	30.6
Full routed system	✓	✓	✓	✓	✓	✓	**73.28**	**19.74**	**0.647**	**0.294**	**0.945**	**33.61**	**29.35**

4.4 Qualitative Analysis

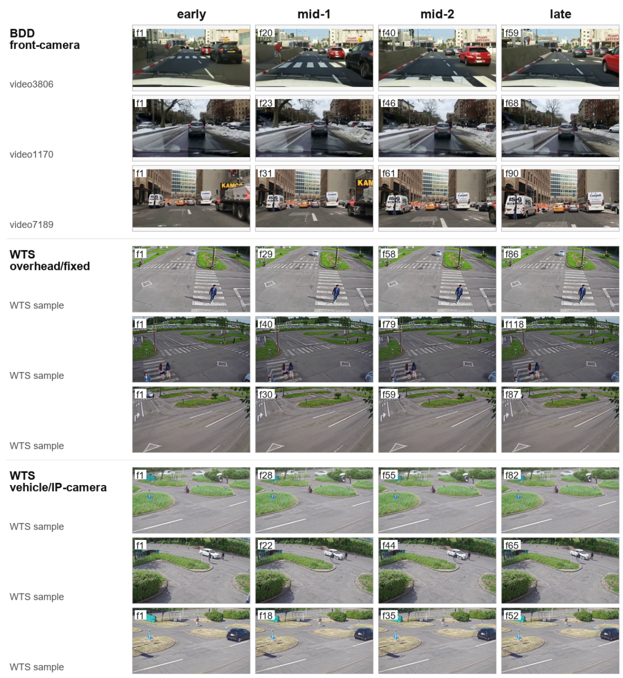

Fig. 5: **Qualitative results across traffic views.** Predicted future frames from representative BDD front-camera, WTS overhead/fixed-camera, and WTS vehicle/IP-camera videos. Each row shows multiple future steps from the same video.

Figure 5 illustrates the behavior of the complete system across heterogeneous viewpoints. Front-camera predictions stabilize the global driving layout, lane boundaries, road geometry, and background structures over long horizons, while actor motion remains inherited from the base generator. For WTS overhead/fixed views, the routed predictor preserves stable backgrounds while propagating localized actor motion. Vehicle/IP-camera examples also avoid the large global distortions that can arise from unconstrained flow propagation.

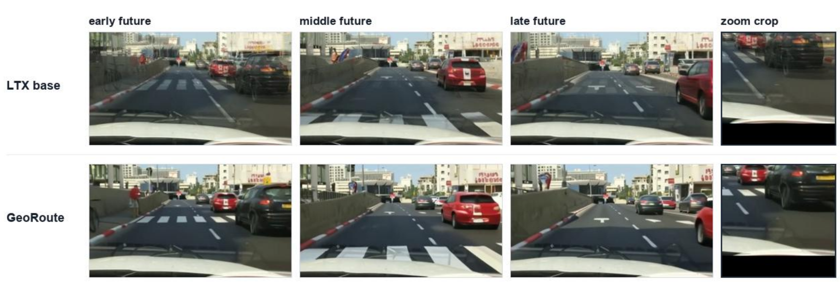

Fig. 6: **Controlled comparison on video3806.** GeoRoute reduces static-region ghosting near lane markings, road boundaries, and actor–background boundaries while preserving dynamic regions from the LTX-Video base prediction [14].

Figure 6 directly compares the base generator with GeoRoute. The LTX-Video [14] sequence blends multiple temporal states around nearby vehicles and road structures, producing visible afterimages. GeoRoute reduces the static component of these artifacts by replacing only reliable background regions with projected history content. Lane markings, curbs, and actor–background boundaries become cleaner, while dynamic and low-confidence regions remain close to the generative prediction. This comparison supports the static-geometry claim but does not imply that GeoRoute re-estimates actor motion.

4.5 Failure Analysis

Geometry refinement depends on relative depth, actor masks, and feature-based alignment. Weak texture, reflections, occlusion, or a poor generated base frame can therefore reject or misalign the projection. In addition, independently canonicalized monocular depth maps do not guarantee a common translation scale for composed multi-history pseudo-transforms. Confidence gating limits the effect of this approximation but does not make the reconstruction metric.

Recent-priority fusion may also select a newer but less accurate source over an older well-aligned frame. Dynamic-motion errors remain in the base generator, and motion propagation may freeze or distort a clip when its observed behavior does not match the selected regime. Finally, the router assumes that the three human-defined regimes remain appropriate; automatic discovery of new regimes is left for future work.

5 Conclusion

In this work, we presented a geometry-aware, training-free framework for traffic future-frame prediction. The method combines a text-conditioned video generator with confidence-aware static refinement and Qwen-assisted routing, improving static-geometry stability and low-level structural fidelity without fine-tuning the pretrained model or changing its sampling schedule. On AI City Challenge Track 5 [36], our system ranks 5th on the official full-test leaderboard with a final score of 73.28. These results suggest that explicit geometric priors and view-aware inference are effective for stabilizing static structure in training-free traffic video prediction. Dynamic-object motion is inherited from the base generator rather than explicitly corrected by the geometry branch. Future work will explore confidence-calibrated learned routing, stronger dynamic-object motion priors, and tighter integration of geometry into video generation models.

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